Yang Liu

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Education

PhD in Biostatistics and Data Science, *University of Texas Health Science Center at Houston* — 08/2021 – 08/2025, Houston, United States
- Relevant Courses: Statistical Inference, Machine Learning, Survival Analysis, Clinical Trials
- Honors: Caroline Ross Fellowship, Julius and Suzan Glickman Endowed Scholarship.

Master in Biomedical Informatics, *University of Texas Health Science Center at Houston* — 01/2019 – 08/2021, Houston, United States
- Relevant Courses: Big Data in Biomedical Informatics, Stat Analysis of Genomic Data
- Honors: Paul C. Boyle Research Award (×2), Dean's Excellence Scholarship, Schull Institute Summer Program Scholarship, James Turley Endowed Scholarship.

Master in Cancer Biology, *Seoul National University* — 08/2015 – 08/2017, Seoul, South Korea
- Relevant Courses: Medical Statistics, Molecular Cell Biology
- Honors: SNU Global Scholarship (×2)

Professional Experience

Graduate Research Assistant, *University of Texas MD Anderson Cancer Center* — 01/2020 – 08/2025, Houston, United States
- **Engineered a novel deep learning framework** combining network-based feature engineering with **graph neural networks (GNNs)** to classify cancer phenotypes and identify biomarkers from gene expression data; secured $519K Moon Shot funding (Briefings in Bioinformatics, 2025)
- **Developed a network-penalized statistical method** for biomarker identification in gene expression datasets with survival endpoints (bioRxiv, 2025)
- **Identified immune biomarkers** in a **phase II glioma trial** of pembrolizumab through gene expression analysis; enabled translational collaboration for treatment response assessment
- **Integrated scRNA-seq and scATAC-seq datasets** to elucidate functional consequences of germline risk variants in glioma progression; awarded $461K R01 funding
- **Enhanced ICD-10 code prediction** from clinical notes by leveraging data-derived knowledge graphs and fine-tuned lightweight large language models (**LLMs**)
- **Developed a RAG tool** to answer graduation-related queries at UTHealth for personalized student support (Huggingface spaces)
- **Built multimodal transformer models** for segmentation of Organ-at-Risk (OAR) regions using paired MRI and CT images (Github)

R&D Engineer, *Berry Genomics* — 11/2017 – 07/2018, Beijing, China
- Developed a BRCA diagnostic kit on NGS platform; optimized primer design for coverage
- Coordinated cross-departmental collaboration for product development

Academic Projects

Microsatellite Status Prediction
Achieved comparable **CNN** performance for classification of histological images with limited GPU resources (Github)

Meta-analysis of mortality rate of COVID-19
Implemented **Bayesian** hierarchical models for mortality estimation

Batch-effect Correction for Single-cell RNA Sequencing Data
Compared 7 methods including Seurat and Harmony for single-cell RNA sequencing data integration

Selected course projects
- Built a **Yelp scraper** to rank nearby Asian restaurants using a custom preference score
- **Imputed missing data** using a custom **EM** algorithm with accuracy matching standard tools.
- Designed a **Markov chain model** to predict outcomes in bone marrow transplant patients
- Developed **optimization algorithms** using coordinate descent and ADMM techniques
- Applied **NLP** methods (word embeddings, random forest) to classify spam emails

Publications & Conferences

- **Authored** 5 first-author publications (3 published, 1 accepted, 1 preprint) including 1 in *Briefings in Bioinformatics*, and 6 co-authored abstracts in *Neuro-Oncology* and *Molecular Cancer Therapeutics*
- **Presented posters** at JSM 2025, SNO Annual Meetings (2021, 2024), and Brain Tumor Center Retreat (2023)
- **Presented at** FY22 Brain Tumor Center Retreat, GraphConnect 2022, and Neo4j AWS GenAI Lab 2024

Skills & Interests

- **Tools:** R, Python, SQL, Neo4j (Cypher), SAS, Tableau, Linux, HPC, Pytorch, CUDA, Version Control (Git)
- **Specializations:** Statistics, AI/ML, Bioinformatics, Bayesian, Simulation, Graph and Image Analysis, LLMs, Wet Lab
- **Languages:** Chinese (Native), English (Proficient), Korean (Proficient)
- **Interests:** Swimming, Hiking, Piano